



Assignment – 3

Design and Analysis of Algorithms (IT2005)
Indian Institute of Information Technology Allahabad

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Instructions

- All answers must be clearly justified. Hand written solution is desirable.
- Algorithms must include time and space complexity analysis.
- Use pseudo-code where appropriate.
- Late submissions will not be consider for evaluation.
- Plagiarism will be checked. AI generated submission will be penalized.

Question 1. Skier–Ski Assignment Problem

Consider the following problem. The input consists of n skiers with heights $p_1, p_2, \dots, p_n$, and n skis with heights $s_1, s_2, \dots, s_n$. The task is to assign each skier exactly one ski so as to minimize the average difference between the height of a skier and the height of the assigned ski. Formally, if skier i is assigned ski $\sigma(i)$ (where σ is a permutation of $\{1, 2, \dots, n\}$), then the objective is to minimize

$$\frac{1}{n} \sum_{i=1}^n |p_i - s_{\sigma(i)}|.$$

Consider the following greedy algorithm:

1. Find the skier and ski whose height difference is minimized.
2. Assign this skier to this ski.
3. Remove both from consideration and repeat the process until every skier has been assigned a ski.

Prove or disprove that this algorithm is correct.

Now consider the following greedy algorithm:

1. Sort the skiers in nondecreasing order of height.
2. Sort the skis in nondecreasing order of height.
3. Assign the shortest skier the shortest ski, the second shortest skier the second shortest ski, and so on.



Prove or disprove that this algorithm is correct.

Hint: One of the above greedy algorithms is correct and the other is incorrect. The proof of correctness must be done using an exchange argument.

Question 2. Job Scheduling with Release Times and Deadlines

We consider two related problems. The input consists of n jobs, where each job j has an integer release time r_j and an integer deadline d_j . In a feasible schedule, each job j must be executed for exactly one unit of time, not starting before r_j and not finishing after d_j . No two jobs may be executed at the same time. Assume all release times are nonnegative, and let

$$T = \max_j d_j.$$

Warm-up Problem. In addition to the above input, for each integer $t \in [0, T)$ it is specified whether a particular machine is turned on during the time interval $(t, t + 1)$. The problem is to determine whether all jobs can feasibly be scheduled for one unit of time during intervals in which the machine is turned on.

Give a greedy algorithm and prove its correctness using an exchange argument.

Main Problem. In addition to the job information, the input contains a positive integer L . The problem is to determine the minimum number k of time intervals of length L such that all jobs can feasibly be executed for one unit of time during one of these k intervals.

An equivalent interpretation is as follows. It costs one dollar to turn the machine on. Once turned on, it remains on for L consecutive time units. The objective is to determine the minimum number of times the machine must be turned on so that all jobs can be completed.

Give a greedy algorithm and prove its correctness using an exchange argument.

Question 3. Gas Station Refueling Strategy

You wish to drive from point A to point B along a highway while minimizing the total time spent stopped for gas. You are given the capacity C of your gas tank (in liters), your fuel consumption rate F (in liters per kilometer), the filling rate r (in liters per minute) at any gas station, and the locations $A = x_1, x_2, \dots, x_n = B$ of the gas stations along the highway.

If you stop to fill your tank from 2 liters to 8 liters, you would spend $\frac{6}{r}$ minutes refueling.

Consider the following two algorithms:

1. Stop at every gas station and fill the tank with just enough gas to reach the next gas station.
2. Stop if and only if you do not have enough gas to reach the next gas station, and if you stop, fill the tank completely.

For each algorithm, either prove or disprove that it correctly solves the problem. Any proof of correctness must use an exchange argument.

Question 4. Minimum Unit Interval Cover

Consider the following problem. The input is a collection $A = \{a_1, a_2, \dots, a_n\}$ of n points on the real line. The goal is to find a minimum-cardinality collection S of unit-length intervals that covers every point in A .

An equivalent interpretation is the following. You are given a collection of times A at which trains arrive at a station. Each arrival time must be covered by an employee. Due to union rules, each employee can work for at most one hour. The problem is to schedule the minimum number of employees so that all arrival times are covered.



- (a) Prove or disprove that the following algorithm correctly solves the problem. Let I be the unit interval that covers the maximum number of points in A . Add I to the solution set S , and recursively continue with the points in A not covered by I .
- (b) Prove or disprove that the following algorithm correctly solves the problem. Let a_j be the smallest (leftmost) point in A . Add the interval $I = (a_j, a_j + 1)$ to the solution set S , and recursively continue with the points in A not covered by I .

Hint: One of the above greedy algorithms is correct and the other is incorrect. The proof of correctness must use an exchange argument.

Question 5. The Integer Knapsack Problem (Duplicate Items Permitted)

You have n types of items, where the i^{th} item type has an integer size s_i and a real value v_i . You are given a knapsack with total capacity C , and your goal is to select items to place into the knapsack such that the total value is maximized. You are allowed to include multiple items of the same type in the knapsack. Determine the maximum total value that can be obtained without exceeding the knapsack capacity C .

Question 6. Box Stacking

You are given a set of n types of rectangular three-dimensional boxes. The i^{th} box has height h_i , width w_i , and depth d_i , where all dimensions are real numbers. You want to build a stack of boxes that is as tall as possible. A box can be placed on top of another box only if both dimensions of the 2-D base of the lower box are strictly larger than the corresponding dimensions of the 2-D base of the upper box. A box may be rotated so that any side can serve as its base, and multiple instances of the same type of box may be used. Determine the maximum possible height of a valid stack of boxes.

Question 7. Optimal Strategy for a Game

Consider a row of n coins with values $v_1, v_2, \dots, v_n$, where n is even. Two players play a game by alternating turns. In each turn, a player selects either the first or the last coin from the row, removes it permanently, and receives the value of the selected coin. Assuming that both players play optimally, determine the maximum possible amount of money that the first player can *guarantee* to win. Compute the maximum value the first player can secure regardless of the opponent's moves.

Question 8. Two-Person Traversal of a Sequence of Cities

You are given an ordered sequence of n cities and the distances between every pair of cities. The task is to partition the cities into two subsequences (not necessarily contiguous) such that:

- Person A visits all cities in the first subsequence in the given order.
- Person B visits all cities in the second subsequence in the given order.
- The sum of the total distances traveled by Person A and Person B is minimized.

Assume that Person A and Person B each start at the first city in their respective subsequences. Determine the optimal partition of cities into two subsequences that minimizes the total distance travelled by both persons.